

OpTM: Analysis of language models vs the state modeling opioid addiction problem

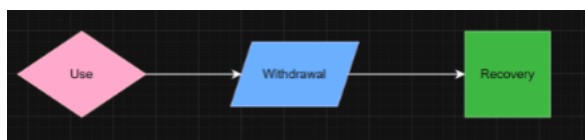
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Abstract

Prior work has studied opioid addiction signals in online communities (MacLean et al., 2015; Jha et al., 2021) and evaluated large language models on medical diagnosis (Qazi et al., 2026; Savage et al., 2024), but less work has examined whether these models can track changing patient states over time. In this paper, we study addiction-state modeling for opioid users on Reddit. We hand-annotate 1754 posts with use, withdrawal, and recovery labels, analyze feature-based neural architectures with CRF transition modeling, and compare them with modern large language models under sliding-window and one-shot prompting. We find that exclusive PMI improves separability and performance in our trained models, while LLMs struggle to distinguish withdrawal from recovery and often fail to produce consistent label sequences as the number of posts increases. These results suggest that longitudinal addiction-state tracking remains difficult for current LLMs, even when the same models perform strongly on many single-prompt reasoning tasks.

1 Introduction



Opioid addiction has devastated communities across the United States on a scale that is difficult to overstate. The CDC estimates that tens of thousands of Americans die from opioid-related overdoses every year, making it one of the leading causes of accidental death in the country. Beyond the death toll, addiction tears apart families. Children are placed in foster care, marriages collapse, and parents watch their kids disappear into a cycle of dependency that is extraordinarily difficult to break. The economic burden is

staggering as well, costing the US hundreds of billions of dollars annually in healthcare, lost productivity, criminal justice involvement, and social services (Luo et al., 2021).

One interesting direction is the determination of addiction states. Why is determining states important? Medicine and treatment also tend to be procedural, so it is useful to break down addiction into components to conduct a more thorough analysis of causality/correlation, compared to treating the user history as a giant blob and throwing it into a classifier. It is also useful because it provides a systematic way to determine *relapse*, otherwise described as a regression in the addiction-recovery cycle.

Learning representations of medical conditions from Internet data is a modern and interesting way to learn about these conditions. Prior work has examined how one can classify mental disorders from social media (Ji et al., 2022), examine words correlated with opioid addiction states from Forum77 (MacLean et al., 2015), analyze information-seeking events in OUD social discourse (Sharif et al., 2024), and evaluate large language models for medical diagnosis (Qazi et al., 2026; Savage et al., 2024).

Particularly, prior work has examined how large language models such as ChatGPT-4 perform on medical diagnosis. The vast majority of the cases that are tested are single-sitting cases, where all the details are placed in a single prompt and the LLM is given a verdict. Indeed, recent work has shown promising results in LLM diagnostic reasoning, including 92% performance in a GPT-4o diagnostic-reasoning setting (Qazi et al., 2026) and diagnostic-reasoning prompt analyses with GPT-4 (Savage et al., 2024).

However, addiction states are particularly interesting. Due to the nature of the data, we have a collection of posts. Then, each post is assigned a label. We then track the posts over time. This

is distinct from much work that examines mental health states, for example, by taking the entire post history and outputting a K-class label (Ji et al., 2022).

Several prior works have analyzed addiction states. Particularly, MacLean et al. collected data from Forum77 to determine the states (MacLean et al., 2015). Jha et al. utilized novel architectures to significantly outperform the preliminary work done in MacLean et al. (Jha et al., 2021).

In this problem, we answer the following questions:

- RQ1: Given a set of posts P from a user u , can we assign a state $s \in [\text{use}, \text{withdrawal}, \text{recovery}]$ to each $p \in P$ that accurately describes their state transitions?
- RQ2: What learning features best contribute to the state classification problem, and how do current large language models perform on time-series diagnostic tasks?

2 Design Analysis

2.1 Dataset Collection

The first analysis we do is cross-location analysis. Prior work examines a few specific online communities and forums:

- Forum77: a general online medical help forum used in MacLean et al. (MacLean et al., 2015)
- r/opiates: a general Reddit forum to discuss opioid use. The posts here are typically recreational.
- r/OpioidRecovery: A subreddit for users discussing their usage in a context of recovery.

One interesting observation when annotating data from a few other subreddits is that certain subreddits share significantly different distributions of classes.

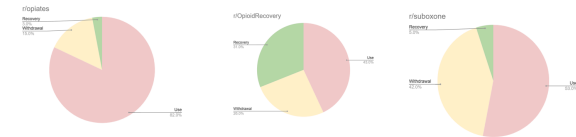


Figure 1: Pie charts of class distribution. Left is r/opiates, middle is r/OpioidRecovery, and right is r/suboxone, a substitute-drug community for withdrawal support. These demonstrate significant class imbalance, as we can see the main Reddit community is mostly use posts. By selectively sampling certain subreddits, we can address the class imbalance in the state labeling

One researcher hand-annotated 1754 posts collected using a Reddit dump (Arctic Shift), placing each post into a class. The overall distribution was 70.4% use, 17.5% withdrawal, and 12.1% recovery. These posts were distributed as follows:

- 1731 posts from r/opiates + r/OpioidRecovery
- 23 posts from r/kratom and r/suboxone

2.2 Data Stability Analysis

The annotation rubric utilizes a class-based annotation taxonomy:

State	Description	Example
[USE]	Subject is using substances and demonstrates no intention to quit	I have tried many ways of potentiating opiates, and mixing them with benzos is by far my favorite. Does anyone else potentiate their opiates? Feel free to offer insight/tips.
[USE] Addiction	Subject is using describes they are addicted, but demonstrates no intent to quit	I've come to my family doctor with a dhc addiction for the last 8 months. I wasn't sure what he would recommend but all he said was just stop and grin and bear it and here's some valium. If it was as simple as just stopping I would have done it but I get horrendously ill. I was hoping for a week taper with suboxone or something
[USE] Relapse	Subject has used substances again after an attempt to quit.	Just wanna say, i've been forced off opiates for about a month now and just recently ran into some fentanyl.
[USE] Intent to quit	Subject expresses desire to stop abusing a substance in the future.	So kinda over did it with my binge. Ended up lating 2 months. Got up to 150+ oxy a day. I need to stop doing this.
[USE] Plan to quit	Subject notes time and/or plan (e.g., tapering schedule) to quit.	So what's the strongest kratom at most smoke shops? Looking to fight w/d and possibly get high off of it after w/d please post the strongest kratom uve seen and what kind. Powder? Capsules? Thanks. Had
[WITHD]	Subject is in withdrawal; method unspecified.	This is hard as hell, it's worse than the first 3 days of withdrawal. Cravings are getting out of hand
[WITHD] Taper	Subject is in withdrawal; detoxification method is a taper.	Things got out of hand pretty quickly and I realized I had to do something so I tapered my habit down to about half a point of (china white?) a da
[WITHD] Cold Turkey	Subject is in withdrawal; detoxification method is cold turkey.	So my fiance and I are quitting right now, her cold turkey (God bless her, I love that woman), and me sub assisted (we share this account, she made a post the other day). We had one sub that we had opened to use, but never took (long story).
[WITHD] Subs	Subject is in withdrawal; is using substitute drugs to help. Mentions both withdrawals and subs.	Been tapering off Suboxone which I used to help curb withdrawals. I feel fantastic and my veins are starting to heal and return which makes me smile.
[RECOV]	Subject has finished detoxing; no physical withdrawal symptoms expressed.	Chalk off another one today pals! Day 78 off the dope for me. Life is better than I ever imagined it could be at the moment. I am going to look at the Turner and Constable paintings at Tate Britain this afternoon. How you all doing?
[RECOV] Maintain	Subject has recovered but is thinking about use.	It's been 152 days (5 months) since I've had any heroin and I still wake up thinking I have to go find a way to score.

2.3 Learning Features and Embeddings

One challenging area is selecting features to learn. The first way is to select word flags that are known to help the model. For example, the word “suboxone”, which is a substitute drug used for tapering, is correlated with withdrawal/use. Another option is to use phrases such as “days clean”, which imply days since a user last used a drug and are strong indicators of recovery.

We test the following methods:

- Pointwise Mutual Information
- Exclusive PMI
- TF-ICF

Pointwise mutual information is a method for determining how important a word is to a class. Given a class C and posts P , the PMI of a class C is defined as:

$$\text{PMI}(\text{word}, \text{state}) = \log \left(\frac{p(s, n)}{p(s)p(n)} \right)$$

For example,

- $p(\text{use}, \text{bob})$ is

$$\frac{[\text{bob}] \text{ tokens in posts with class } [\text{use}]}{\text{all tokens}}$$

If [bob] is highly associated with the class, then most [bob]s will appear in that class. Hence, the PMI score will be greater. The converse is true.

Of course, this is skewed by tokens that appear more rarely, such as the word “upvote”. If $p(n)$ is generally small, then the score will increase rapidly. However, we found here that the words discovered by setting a minimum frequency, e.g., a word has to appear 25 times to be considered, are more useful since rare words in general are not that useful.

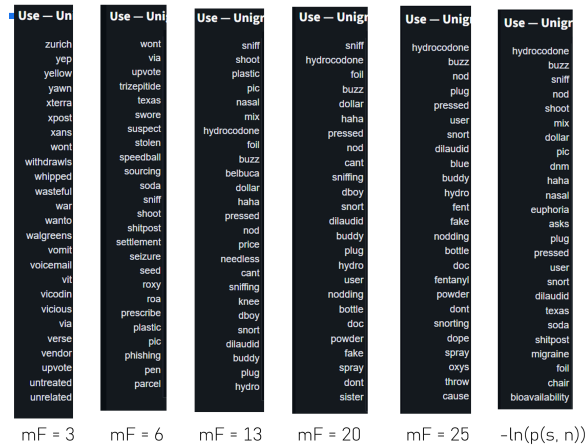


Figure 2: Word list discovered by PMI at various minimum frequencies. Setting minimum frequency to 25 results in slight differences in the found words. With the normalization strategy, weaker words such as “shitpost”, “texas”, and “mix” still appear, whereas minimum-frequency filtering outright disallows rarer words, giving a much more condensed list. However, one other interesting direction is the concept of “exclusivity”. What words are much more associated with a state? To account for this, we add an exclusive weighting to PMI. This penalizes tokens severely for appearing in the other classes. Essentially, we have:

$$\text{EPMI}(\text{word}, \text{state}) = \text{PMI}(s, n) - \sum_{i \notin \text{state}} \text{PMI}(i, n)$$

The subtracted term is a sum of PMI of the other states. For example, if we are finding it for [use],

the subtracted term is the sum of the PMI for [withdrawal] and [recovery].

TF-IDF. We considered using TF-IDF. However, the results were generally noisy. TF-IDF does not provide any penalization for documents appearing across classes, hence most of the words it found were not useful.

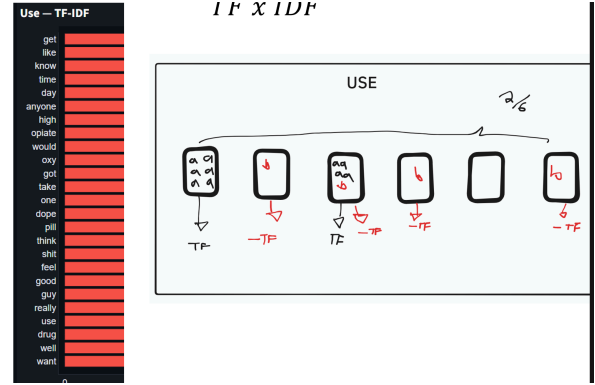


Figure 3: TF-IDF word list and diagram. Even with stop-list removal, stemming, and lemmatization, we could not get a good word list. Generally, these words may have useful importance, but they are too shared between classes to be useful.

As an alternative, we used TF-ICF, where the ICF term serves as inverse class frequency. This is defined as $\log \left(\frac{\text{number of classes}}{\text{classes that contain token}} \right)$. This way, the term is applied across documents.

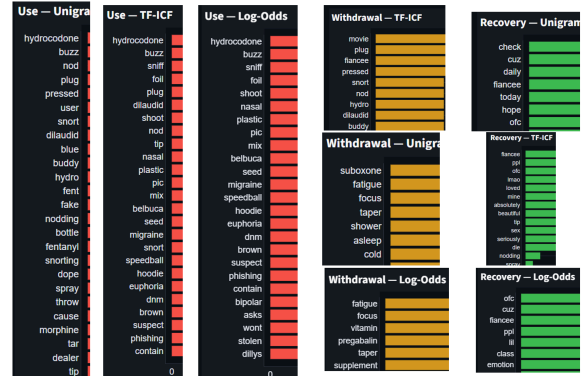


Figure 4: PMI list with minimum frequency 25, EPMI, and TF-ICF. The word lists generated with EPMI and TF-ICF look worse, however, it is likely they may help training. The concern is overfitting, if certain words have exclusivity just because of the training data.

For embeddings, we opted to choose word2vec. Word2vec was used to generate a 100-dimensional embedding for each word. The post is then the centroid of the word embeddings.

We tried to use a tokenization strategy, which unlike Word2Vec preserves word ordering. However, the sequences for posts are very long and result in very large models. Generally, the

amount of posts is insufficient for this strategy, even when using a pretrained BERT tokenizer.

2.4 Learning Architecture and Challenges

After converting our features to vectors, we examine a few interesting directions of deep learning. The first one is the LSTM. This is a gated neural network that provides better memory than RNNs. Recent advances have shown interesting directions for transformer-based language models, for example BERT. The general architecture follows this diagram:

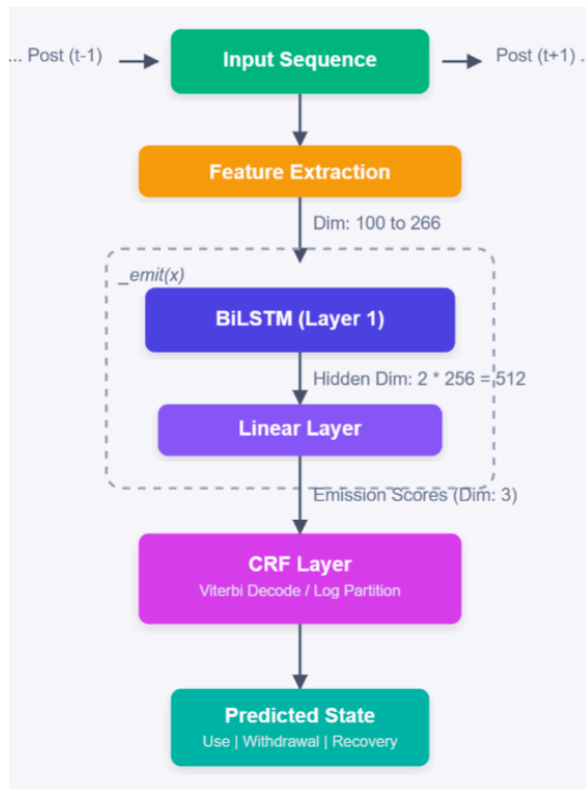


Figure 5: Diagram of the model architecture. The initial post vector is placed into the model, which has a linear layer that outputs an emission score. These emission scores are placed into the CRF layer, which then outputs the predicted state

Thus, our architectures were:

- Dense model, directly feeding the 100-dimensional post vector.
- State-specific, 3 embedding layers each with state PMI word2vec of size 50. All words not in PMI are purged.
- Sequence: Tokenize all words with a tokenizer, train on left-padded sequences.
- Window: Add the previous 2 posts generically, no PMI, as context and the 2 next posts directly into the BiLSTM. Apply CRF on top.

However, the first concern is the amount of data. Due to the parameter counts of these large models, it is a major concern. Thus, we try finetuning large models, since prior work has shown they are useful with 1000-2000 examples. An important thing to notice is the overfitting problem. When working with large architectures, most models overfit.

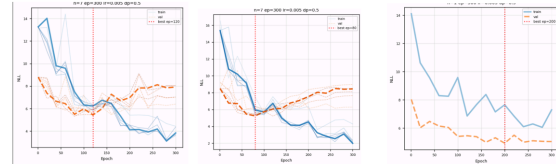


Figure 6: The validation loss vs. training loss for 900 pieces of data. Model 1 was a large LSTM, model 2 was a ModernBERT finetune, while model 3 was a smaller model. Larger models had a tendency to overfit, while model 3 was relatively underfit but performed the best on validation.

To get around this problem, we expanded the dataset to 1754 annotations to examine how larger language architecture performance scaled with data. However, even with the larger dataset, finetuning BERT demonstrated strong overfitting and weak performance.

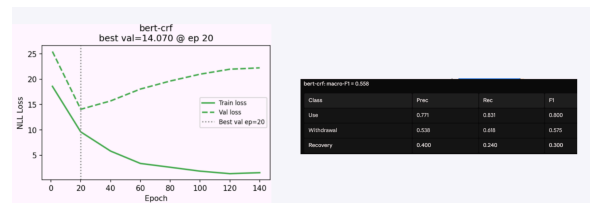


Figure 7: Training loss vs validation loss graph. F1 table. Here we can see that BERT overfits early on in training. Even when using this as the best checkpoint, the BERT model capped at 0.3 F1 for recovery.

Additionally, we utilize a conditional random field to learn the transitions. CRFs are commonly used in prior work and in recent sequence-labeling systems (OpenAI, 2026b). It makes sense to use a CRF here because state transitions are constrained; it does not make much sense for a user to jump between states rapidly, for example.



Figure 8: A user's state over time. The vast majority of the user's state stays as one class, with occasional jumps. There is a little fluctuation in states but not significant fluctuation (e.g., 4+ transitions in a fixed time).

2.5 Large Language Models

Transformer architectures and large language models have emerged recently as prominent systems for medical tasks. Several prior works have examined their use in medical diagnostic reasoning. In this paper, we examine the following models:

- Qwen3.6-27B. This is a smaller model released by Alibaba that demonstrates strong performance in many coding and reasoning tasks (Qwen Team, 2026).
- Gemini 3.1 Pro. This is a large model used in Google’s Gemini service (Google Gemini Team, 2026).
- GPT-5.5 Thinking. This is a frontier LLM from OpenAI released in 2026 (OpenAI, 2026a).
- Claude Sonnet 4.6. This is another frontier model from Anthropic, featuring strong coding performance (Anthropic, 2026).

Due to the size of our data, there is a high interest in the use of large language models due to their promising ability in zero-shot reasoning tasks.

2.5.1 The LLM workflow

One challenging aspect of large language models is constrained decoding. In our case, when given N posts, we must have N labels. What often happens is the LLMs do not consistently output N labels, particularly when the number of posts given to the large language model scales. We utilize two prompting frameworks:

- One-shot all posts. Every post is included in the prompt, and the LLM is to output the list of labels.
- Sliding window of $1-K$ posts, moving 1 post to label at a time. The previous 3 posts are attached to serve as context.

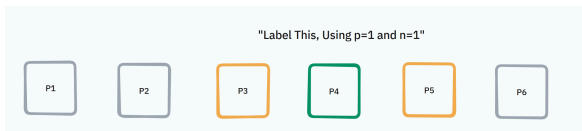


Figure 9: The sliding window architecture. The green is the post being labeled. Yellow are posts included in the context that won’t be labeled. Gray posts are omitted. Only the next post and previous post are included because $p = 1$ and $n = 1$. If $p = 2$ for example, then Post2 would have been included in the context. If we expand the labeling window by 1, P5 would be green, and P6 would be yellow.

We do a robustness check of the sliding window approach. The first issue with the sliding window is determining which posts are added. We initially tried with the 3 posts previous to the posts currently being labeled. However, this resulted in issues where the model could not uniquely determine the forward context.

But interestingly, adding in the future context window did not improve performance. In fact, adding 3 posts from the future resulted in no significant difference.

	Accuracy	Stdev
0 posts	62%	4.20%
last 3 posts	72%	2%
prev 3 + next 3	68%	1.60%

Figure 10: Five runs using Qwen3.6-27B, thinking enabled. The p-value was 0.73, demonstrating no significant difference. Both context settings were significantly different from 0 posts.

We decided to use simply the last 3 posts with one post being labeled.

2.5.2 Feasibility of one-shot all posts

One-shotting all the posts provides interesting performance. Specifically, much prior work demonstrates that new large language models can benefit from more context. So by inputting more context, we can improve their performance. Indeed, running it on the validation set results in good performance.

	Accuracy	Stdev
last 3 posts	72%	2%
one-shot	83%	3%

Figure 11: One-shot with Qwen3.6-27B performed significantly higher than the SOTA last-3-posts setting. The p-value was 0.04, indicating statistical significance.

However, the feasibility of these models lies in their ability to consistently generate a list of size n , where n is the number of input posts.

	Qwen3.6 27B	Gemini 3.1 Pro	GPT5.5 Thinking	Sonnet 4.6
10 posts	100%	100%	100%	100%
20 posts	95%	100%	100%	100%
30 posts	76%	86%	83%	86%
50 posts	32%	46%	48%	52%
100 posts	21%	29%	28%	31%

Figure 12: LLM consistency across post lengths. When given many posts, LLMs all have difficulty generating the required number of labels.

What is particularly interesting with LLM scaling is that larger parameter counts did not help performance in our experiments. We include the

table of large language model performance for discussion:

	Accuracy
Qwen3.6 27B	80%
GPT5.5 Thinking	78%
Gemini 3.1 Pro	83%
Claude Sonnet 4.6	81%

Figure 13: Accuracy of modern large language models on this task. Each model performs within the others’ range, showing no significant difference. In our experiments, we did not observe a clear performance improvement from larger parameter counts. In-depth analysis is provided in results.

3 Results

3.1 F1 Evaluation of Training Models

Use				Withdrawal			
Method	Precision	Recall	F1	Method	Precision	Recall	F1
Dense	0.935	0.892	0.913	Dense	0.750	0.353	0.480
State-Specific	0.822	0.923	0.870	State-Specific	0.778	0.412	0.538
SeqModel	0.981	0.815	0.891	SeqModel	0.600	0.618	0.609
Window+CRF (k=3)	0.697	0.954	0.805	Window+CRF (k=3)	0.538	0.206	0.298

Recovery			
Method	Precision	Recall	F1
Dense	0.500	0.920	0.648
State-Specific	0.545	0.720	0.621
SeqModel	0.690	0.840	0.760
Window+CRF (k=3)	0.591	0.520	0.553

Figure 14: Recall and precision scores of the LSTM. We see here that we achieve excellent recall for use. However, it is also the case that the architectures trade blows. In particular, the dense and sequence models have very high recall on recovery and withdrawal. Sequence and windowCRF do not use PMI, which may be why their withdrawal precision is much lower. The use of PMI lets dense and state-specific models achieve good withdrawal precision, but much worse recall.

Use				Withdrawal			
Vocab	Precision	Recall	F1	Vocab	Precision	Recall	F1
PMI	0.818	0.831	0.824	PMI	0.727	0.235	0.356
log-PMI	0.910	0.938	0.924	log-PMI	0.750	0.706	0.727
TF-ICF	0.859	0.938	0.897	TF-ICF	0.810	0.500	0.618

Recovery			
Vocab	Precision	Recall	F1
PMI	0.404	0.760	0.528
log-PMI	0.760	0.760	0.760
TF-ICF	0.594	0.760	0.667

Figure 15: Table of F1 performance difference between learning features on the state-specific model. EXL-PMI performed strongly overall. TF-ICF performed generally worse, and PMI-Only performed worse than EXL-PMI. This was using the best model from 4 runs each. The p-value from Mann-Whitney U was 0.032 for EXL-PMI vs PMI-only. We decided that exclusive PMI was worth using.

3.2 Model Training and Area under the Curve

We examined the ROC and AUC for our architectures:

Model	Use	Withdrawal	Recovery
dense	0.949	0.649	0.893
state-specific	0.946	0.732	0.870
seqmodel	0.957	0.770	0.881
ctx3-window	0.946	0.759	0.864

Figure 16: The AUC for use is very high. We see that all models have a significantly harder time determining the boundary between withdrawal and recovery. This is likely due to an interesting set of posts in the test set mentioning “days clean”, but they are still in withdrawal because they experience strong symptoms.

3.3 Confusion Matrix and Error Clustering

Here, we examine the area under the curve, the confusion matrix, and semantic clustering of the errors.

We construct the confusion matrix of several models.



Figure 17: Confusion matrix for the architecture models. Darker regions mean higher prediction rates. We can see that all models have severe difficulty classifying withdrawal, as it is relatively spread out. The boundary between recovery and use is high, but models commonly predict recovery as withdrawal. Overall, withdrawal suffers from the highest misclassification. This makes sense, likely due to the fact that withdrawal is the state between the other states.



Figure 18: Exclusive selection affects separability. We can see that on the leftmost chart, the middle of the chart for withdrawals is light. The rightmost chart (TF-ICF) has strong separability, but commonly confuses recovery and withdrawal. However, exclusive PMI strongly separates the classes, leaving all non-diagonal elements light.

Semantic clustering of the errors. These errors had embeddings generated via MiniLMv6 and were clustered using DBSCAN. An LLM annotated the top 5 posts of each cluster.

Cluster	Posts	Top error types	Sample text
Cluster 0	34	Recovery-vic13, Withdrawal-vic13, Use-Recovery-7	Sunday October 30th checking in. Ugh. Woke up feeling like shit. Got two hours of sleep cause of the damn rh. Start my...
Cluster 1	20	Withdrawal-vic17, Withdrawal-Recovery-3	Poor pain?? I've accepted that I'm not gonna sleep well, but my best, specifically my heels, are in so much pain right...
Cluster 2	7	Recovery-vic4, Withdrawal-vic4, Withdrawal-Recovery-1	Salvation Army ans. I just left an apt in Northern California and am just wondering if anyone else has had experience...
Cluster 3	5	Recovery-vic12, Withdrawal-Recovery-2, Recovery-Withdrawal-1	60 days. Just realized today is my 60 days. Life is good I've got a job, supportive family, and a roof over my head...
Cluster 4	36	Withdrawal-Recovery-32, Recovery-vic12, Recovery-Withdrawal-1	Friday Nov 4th check in. Day 10. Double digit. Passed a drug test this morning for the first time in months. Last night...
Cluster 5	11	Use-Withdrawal-1, Use-Recovery-3	je t'aime update. comment? "body" "we were" "created at" 14932224416, "score" 4, "author" "YourCardinal", "c...
Cluster 6	12	Withdrawal-vic14, Withdrawal-Recovery-15, Recovery-vic14	Whisperer group. So I just got home two months ago thanks to the advice that and the feeling that people for it...
Cluster 7	10	Recovery-vic1, Withdrawal-Recovery-3, Use-Withdrawal-2	Wanted and didn't. So I just got a washed shoe. My second one. And the doctor told me that wasn't best choice from...
Cluster 8	13	Withdrawal-vic14, Recovery-vic14	Medical situation. So there's a subconscious trigger and need to have it taken out. It's making me suicidal and not here...
Cluster 9	9	Recovery-vic4, Withdrawal-vic13, Withdrawal-Recovery-2	2 months. What music helped you in recovery. For me there's a lot of music that's currently getting me through sobriety.

Figure 19: The semantic clustering of the errors. The most common errors were withdrawals being misclassified as recovery, and vice versa. Many of the misclassifications mention days, which is interesting. A lot of substitute drugs generally get marked as use.

3.4 Correlation between LSTM errors and Large Language Models

We analyze the performance of large language models.

Withdrawal				Use			
Run	Precision	Recall	F1	Run	Precision	Recall	F1
aft-cntx	0.636	0.412	0.500	aft-cntx	0.942	0.754	0.838
no-aft-cntx	0.773	0.500	0.607	no-aft-cntx	0.961	0.754	0.845
no-aft-cntx2	0.750	0.441	0.556	no-aft-cntx2	0.961	0.754	0.845
onshot-gemini3.1	0.842	0.471	0.604	onshot-gemini3.1	0.954	0.954	0.954
onshot-gpt5.5	0.778	0.412	0.538	onshot-gpt5.5	0.967	0.892	0.928
onshot-qwen	0.750	0.529	0.621	onshot-qwen	0.967	0.892	0.928

Recovery			
Run	Precision	Recall	F1
aft-cntx	0.480	0.960	0.640
no-aft-cntx	0.451	0.920	0.605
no-aft-cntx2	0.453	0.960	0.615
onshot-gemini3.1	0.625	1.000	0.769
onshot-gpt5.5	0.543	1.000	0.704
onshot-qwen	0.575	0.920	0.708



Figure 20: LLM performance and confusion matrix. LLMs have strong recall for recovery, strong precision for use, and strong precision for withdrawal. However, they also have false positives. Compared with the trained architectures, their confusion matrices look broadly similar, except for the state-specific EPMI model. Recoveries are constantly confused with withdrawal. When LLMs cannot immediately see the future posts, they suffer more from false positives compared to base LLMs. This suggests that LLM architectures also have severe issues distinguishing between withdrawal and recovery.



Figure 21: Embedding space of errors. We see that the LLM errors are spread similarly to the other models. We recommend going to the website to interactively look at the error space.

Generally, large language models suffered from distinguishing withdrawal and recovery.

4 Discussion & Conclusion

It is an interesting problem. Large language models show promising capabilities in labeling the task. However, they struggle with the state-tracking problem over time, performing worse than CRF models despite having much stronger general reasoning capabilities. A key issue is how eager LLMs are to jump. Particularly when looking at LLMs through the next-post approach

strategy, LLMs achieve significantly lower results than expected.

Even when looking at entire posts, LLMs demonstrate limited abilities. One particularly interesting aspect is that it is easy to point at any post later in a chat and convince the LLM that it got it wrong. Nonetheless, even the latest large language models struggle with this task, which is interesting. Prior work has examined how LLMs can label mental health problems when given the entire user history. However, little work has examined their use in state tracking over time. This paper examines state modeling with LLMs in a constrained environment where there is both prior literature on how older architectures performed, and very definite states that change with large Internet data due to the nature of Reddit.

We conclude about the possibilities of addiction state labeling, but also review modern architectures. The answer to RQ1 is the same as with prior work in that we can characterize and identify users in addiction states. The answer to RQ2 is a lot more interesting. We have analyzed how PMI embeddings and exclusivity can help distinguish these blurry states. We have shown how even large language models struggle to distinguish the states. The particularly interesting thing is that exclusive PMI improves separability and performance in our trained models.

5 Data

We include the dataset in this paper:

- LLM Responses
- The OpTM Dataset
- Pretrained pytorch models.

More interactive exploration can be done here: <https://da.mzen.dev>

References

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